Q2. Observation and explanation



Figure 1. Two provided filters applied to the original image

Observation: Filter 1 gives a washed-out, mostly white version of the original, with the colors still faintly visible and the edges only slightly darker, so it does not isolate edges. Filter 2 gives a mostly black image with thin bright lines along the veins, spots and wing outline, so it detects edges.

Reason: Both kernels measure the difference between the center pixel and its neighbors, but their weights sum differently. A kernel acts as an edge detector only when its weights sum to zero. Filter 2 does, and Filter 1 does not.

- Filter 1: the weights sum to $4 - 8 = -4$. In flat regions of intensity v, the output is about $-4v$ in each channel. After taking the absolute value and clipping to 255, every bright region (the yellow wing, the pink flowers) saturates toward white. Only the very dark veins stay darker, and the leftover color tint comes from the channels that did not fully saturate.
- Filter 2: the weights sum to $4 - 4 = 0$. In flat regions the output is about 0 (black). At edges, the center differs from its diagonal neighbors, so the output is large (bright). It is a high-pass Laplacian-type filter, and because it uses only the diagonal neighbors, it gives thin lines.